Determinants of Depression among University Students: An Econometric Analysis using Logit and Probit Models

Report submitted to
Dr. Prantik Bagchi
Department of Humanities and Social Sciences
Indian Institute of Technology Kharagpur

Submitted by

Atharv Lodhi
Sourav Choudhary
Piyush Bisht

Department of Humanities and Social Sciences
Indian Institute of Technology Kharagpur
Kharagpur, West Bengal 721302

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Atharv Lodhi
Dept. of Humanities and Social Sciences
IIT Kharagpur

Sourav Choudhary
Dept. of Humanities and Social Sciences
IIT Kharagpur

Piyush Bisht
Dept. of Humanities and Social Sciences
IIT Kharagpur

***Abstract*—Depression is one of the most common mental health problems among university students, yet it is unclear which personal, social and economic characteristics separate depressed students from others once they are considered together. Using survey data on 300 university students aged 18–24, of whom 89 (29.7%) report depression, this paper estimates binary logit and probit models with average marginal effects, supported by chi-square and t-tests, multicollinearity, goodness-of-fit and robustness checks. Being single is the strongest risk factor (+36.4 percentage points, $p < 0.001$). A rich family background (-17.2 percentage points) and ever being married (-16.4 percentage points) are protective, while each additional thousand units of monthly living cost raises the probability by 1.15 points. Male students are 10.7 points less likely to be depressed than female students. CGPA, semester, age, living arrangement and having a friend circle show no significant effect. The logit model is jointly significant (LR $\chi^2(11) = 69.57$, $p < 0.001$), well calibrated (Hosmer–Lemeshow p = 0.314) and has an AUC of 0.781; probit and linear probability models give almost identical effects. The results suggest that student mental health support should prioritise social connection and financial pressure rather than academic performance alone.**

***Index Terms*—Depression, university students, binary logit, probit, marginal effects, relationship status, socioeconomic status, mental health.**

I. INTRODUCTION

A. Background

University life combines several major transitions within a few years: students move away from home, form new friendships and romantic relationships, manage their own money and face continuous academic evaluation. For many students these changes are a source of persistent stress, and depression—marked by low mood, loss of interest, fatigue and feelings of worthlessness—is among the most frequent mental health problems in this group. A systematic review of university students reported an average depression prevalence of about 30%, well above the general population [1]. Student depression also lowers academic productivity and raises the risk of dropping out [2], [3], so knowing which students are most at risk matters to universities deciding where to direct limited counselling resources.

Theory points to many interrelated causes: stressful life events acting on individual vulnerability [4], the protective role of social support [5], a socioeconomic gradient in depression [6] and higher depression among women [7]. This paper brings these strands together and tests them jointly on one well-defined group of students.

B. Problem Statement

Many studies examine one factor at a time—for example grades or family income—and therefore cannot tell whether an apparent effect is real or reflects other, correlated characteristics. If single students are also more likely to live away from family, a simple comparison cannot separate the two. There is also a widespread assumption that poor grades are a main driver of depression; if this is wrong, support programmes focused on academic help will miss many students in need. This study estimates multivariate probability models that isolate the contribution of each factor while holding the others constant.

C. Objectives and Research Questions

The study aims to (i) measure the prevalence of depression in the sample; (ii) compare depression across gender, family background, relationship and marital status, living style, social support and health; (iii) estimate the effect of each factor with logit and probit models; (iv) express the effects as changes in probability; (v) test their robustness; and (vi) derive recommendations for universities. It asks whether academic performance, relationship and marital status, family background and living costs, gender, and social support are associated with the probability of depression.

D. Hypotheses

Based on the literature in Section II, the following hypotheses are tested:

- **H1:** Students with higher CGPA are less likely to be depressed (academic stress).
- **H2:** Students with a friend circle are less likely to be depressed (social support).
- **H3:** Single students are more likely to be depressed than students in a relationship.
- **H4a:** Students from lower-middle class families are more likely to be depressed than middle class.
- **H4b:** Students from rich families are less likely to be depressed than middle class.
- **H5:** Higher monthly living cost is associated with higher depression (financial pressure).
- **H6:** Students living with family are less likely to be depressed than those living as bachelors.
- **H7:** Female students are more likely to be depressed than male students.

E. Contributions and Organization

The paper contributes evidence on the relative importance of social, economic and academic factors measured on the same students, reports marginal effects that translate results into probability changes, and offers practical guidance for student support services. Section II reviews the literature, Section III describes the data and methodology, Section IV presents the results, Section V discusses them, Section VI covers limitations and future research, and Section VII concludes. The appendix reports the full model output.

II. LITERATURE REVIEW

A. The Diathesis–Stress Model

The diathesis–stress model holds that depression develops when stressful events act on a pre-existing vulnerability [4]. Students face a dense cluster of stressors—examinations, financial obligations, relationship changes and living away from home—so the model predicts higher depression among students exposed to more stressors or with fewer buffers. Beck's cognitive theory adds that negative interpretations of such events turn stress into depressive symptoms [8].

B. Social Support and the Stress-Buffering Hypothesis

Social support protects mental health mainly by buffering the impact of stress: people with close, supportive relationships cope better with the same adversity [5]. Friendships, partners and family reduce the risk of depression through emotional support and a sense of belonging [9]. For students, friend circles, partners and living with family are the most direct sources of such support (H2, H3, H6).

C. Academic Pressure and Performance

Academic stress is widely cited as a driver of student depression, and depression is associated with lower grade point averages [2], [3]. The direction of causality is unclear, however, and links often weaken once social and economic variables are included. H1 tests whether CGPA has an independent association.

D. Socioeconomic Status and Financial Strain

People with lower socioeconomic status have significantly higher odds of depression, with financial strain a main channel [6]. Debt and money worries are strongly linked to poor mental health [10]. Family background shapes both resources and expectations, while monthly living cost directly measures financial pressure (H4a, H4b, H5).

E. Relationships, Marriage and Gender

The quality and stability of romantic relationships are closely tied to depressive symptoms [11], and the link between marital status and depression runs in both directions [12]. Being single can bring loneliness in early adulthood (H3). Women experience depression roughly twice as often as men [7], [13], a gap attributed to differences in stress exposure, coping styles and biology (H7).

F. Physical Health and Summary

Chronic physical conditions sharply increase the probability of depression [14], so self-reported health is expected to be strongly related to depression. When a health measure partly reflects the same symptoms, however, it cannot be treated as an independent cause—an issue that proves important here. Overall, the literature identifies academic, social, economic, demographic and health factors, but rarely estimates them jointly in one sample; this paper addresses that gap.

III. DATA AND METHODOLOGY

A. Data

The study uses survey data on university students. The analysis sample consists of 300 students aged 18–24 (144 male, 156 female), drawn by simple random sampling with a fixed seed from the students of one home region in a larger student survey. Using one region removes regional differences that could confound the comparison, and the 18–24 range covers the typical undergraduate age. The region identifier was removed after sampling. For each student the data record gender, age, family background, relationship status, marital status, living style, current semester, CGPA, friend circle, self-reported health, monthly living cost excluding study costs and depression status. Table I shows the first five records.

TABLE I
RAW DATA (FIRST FIVE RECORDS)

Gender	Age	Family background	Relationship	Marital status	Living	Semester	CGPA	Friends	Health	Monthly cost	Depression
Female	23	Middle class	Single	Unmarried	Family	4	2.92	No	Good	7400	No
Male	20	Lower Middle class	Single	Unmarried	Bachelor	5	3.10	Yes	Normal	5896	No
Male	20	Middle class	Single	Unmarried	Bachelor	5	3.59	Yes	Abnormal	14815	Yes
Male	23	Middle class	In a Relationship	Unmarried	Family	2	3.00	Yes	Normal	19681	No
Male	22	Middle class	In a Relationship	Unmarried	Family	3	3.00	Yes	Normal	15750	No

B. Data Transformation

The data contain 0 missing values and 0 duplicate records. Text answers were converted to 0/1 indicators (depression, male, living with family, friend circle). Family background was coded as two dummies (lower-middle class, rich class) with middle class as the base. Two very small categories were merged: the 'complicated' relationship group (11 students, none depressed) would be predicted perfectly, so relationship status became **single vs. not single**; the 'separated' group (6 students) was merged with married into **ever married**. Monthly cost was divided by 1,000.

Self-reported health is almost perfectly aligned with depression: 100% of students in critical health and 0% in good health are depressed. Including it in the main model would cause quasi-complete separation, and it very likely measures the same condition as the outcome. It is therefore analysed descriptively and entered only in one robustness model. Table II describes the final variables.

TABLE II
DESCRIPTION OF VARIABLES

Variable name	Description and coding	Data type
depression	Whether the student reports depression. 1 = Yes, 0 = No.	Binary (dependent variable)
male	Gender of the student. 1 = Male, 0 = Female.	Binary
age	Age of the student in years (18-24).	Numerical (discrete)
cgpa	Cumulative grade point average on a 4.00 scale.	Numerical (continuous)
semester	Semester the student is currently in.	Numerical (discrete)
cost_k	Monthly living cost excluding study costs, in thousands of local currency units.	Numerical (continuous)
fam_lowermid	1 = Lower-middle class family (base group: middle class).	Binary (dummy)
fam_rich	1 = Rich class family (base group: middle class).	Binary (dummy)
single	1 = Single, 0 = In a relationship or complicated.	Binary
ever_married	1 = Married or separated, 0 = Unmarried.	Binary
live_family	1 = Lives with family, 0 = Lives as a bachelor (hostel/mess/shared).	Binary
friend_circle	1 = Has a friend circle, 0 = Does not.	Binary
health_poor	1 = Abnormal or critical health, 0 = Good or normal (robustness model only).	Binary
health_level	1 = Good, 2 = Normal, 3 = Abnormal, 4 = Critical (descriptive only).	Ordinal

C. Econometric Models

Because the outcome is binary, ordinary least squares can predict values outside 0–1 and is heteroskedastic by construction, so binary choice models are used [15], [16]. The logit model is

$$\Pr(y_i = 1 \mid x_i) = \Lambda(x_i'\beta) = \exp(x_i'\beta) / [1 + \exp(x_i'\beta)] \quad (1)$$

with $x_i'\beta = \beta_0 + \beta_1\text{Male} + \beta_2\text{Age} + \beta_3\text{CGPA} + \beta_4\text{Semester} + \beta_5\text{Cost} + \beta_6\text{LowerMid} + \beta_7\text{Rich} + \beta_8\text{Single} + \beta_9\text{EverMarried} + \beta_{10}\text{LiveFamily} + \beta_{11}\text{FriendCircle}$. Coefficients are estimated by maximum likelihood; $\exp(\beta)$ is the odds ratio. The probit model replaces Λ with the standard normal CDF Φ:

$$\Pr(y_i = 1 \mid x_i) = \Phi(x_i'\beta) \quad (2)$$

Because coefficients of the two models are on different scales, results are reported mainly as average marginal effects (AME): the average change in $\Pr(y = 1)$ when a continuous variable rises by one unit, or when a 0/1 variable changes from 0 to 1 (Stata: margins, dydx(*)).

D. Diagnostics, Robustness and Software

Fit is assessed with the likelihood-ratio test, McFadden's pseudo-R^2, AIC/BIC, classification tables, the area under the ROC curve and the Hosmer–Lemeshow test [17]; multicollinearity with variance inflation factors. Robustness checks use robust standard errors, a linear probability model (LPM), a model adding health status, and separate models for male and female students. Estimation used Python (pandas, SciPy, statsmodels [18]); an equivalent Stata do-file reproduces every step [19]. The data are anonymised, contain no personal identifiers, and results are reported only at group level.

IV. EMPIRICAL RESULTS

A. Descriptive Analysis

Depression is common: 29.7% of students—almost three in ten—report it (Fig. 1), in line with international estimates of around 30% [1]. The outcome is balanced enough for a binary choice model, with enough depressed students relative to the number of regressors for stable estimation.

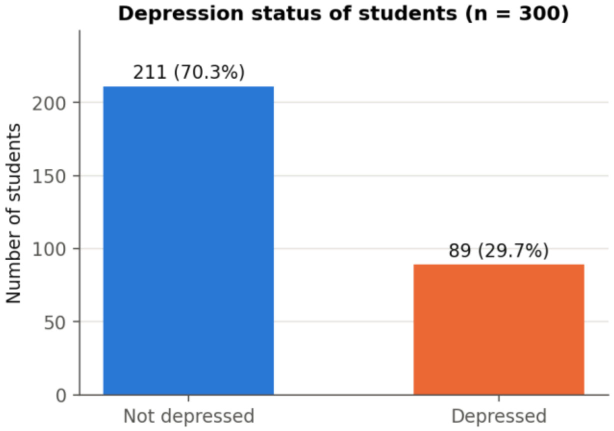


Fig. 1. Distribution of the dependent variable (depression status).

Fig. 2 compares depression rates across groups. Relationship status shows the largest gap: 42.2% of single students are depressed versus 8.2% of students in a relationship and none of the 11 in a complicated relationship. Middle-class students have the highest rate (35.8%), above lower-middle class (19.3%) and rich students (18.0%). Students living as bachelors are more often depressed (33.7%) than those living with family (23.3%). Gender and friend circle show small raw differences (male 31.9%, female 27.6%; friend circle 30.0% vs. 28.7%). These raw comparisons ignore overlaps between characteristics—single students are more often male and less often live with family—which the regression models separate.

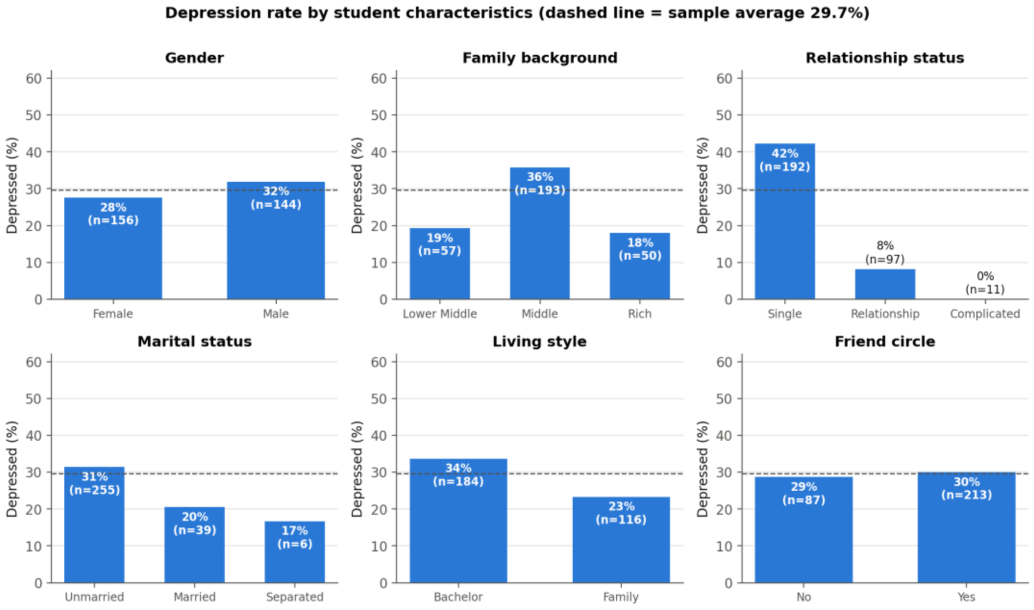


Fig. 2. Depression rate by student characteristics (dashed line: sample average).

Fig. 3 shows how closely self-reported health tracks depression: none of the students in good health and only 2.9% in normal health are depressed, against 59.4% in abnormal and 100% in critical health. Health therefore nearly predicts the outcome and is not used as a regressor in the main model.

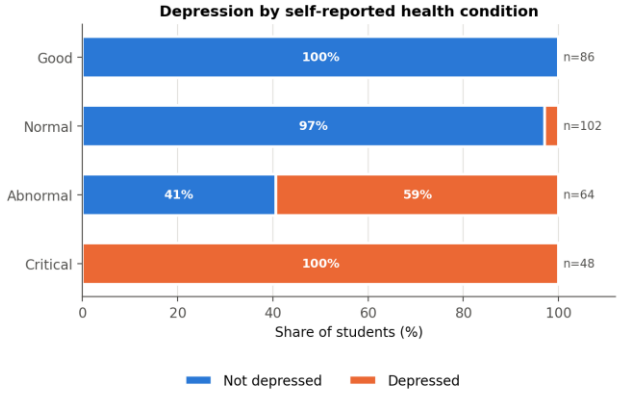


Fig. 3. Depression by self-reported health condition.

Fig. 4 compares the numerical variables. Mean CGPA (3.48 vs. 3.51) and age (21.7 vs. 21.7) are almost identical for non-depressed and depressed students, while depressed students report a somewhat higher monthly cost (14.43 vs. 12.96 thousand). Depression rates by age and semester (Fig. 5) show no clear trend; the very small groups at ages 18 and 19 should be read with caution.

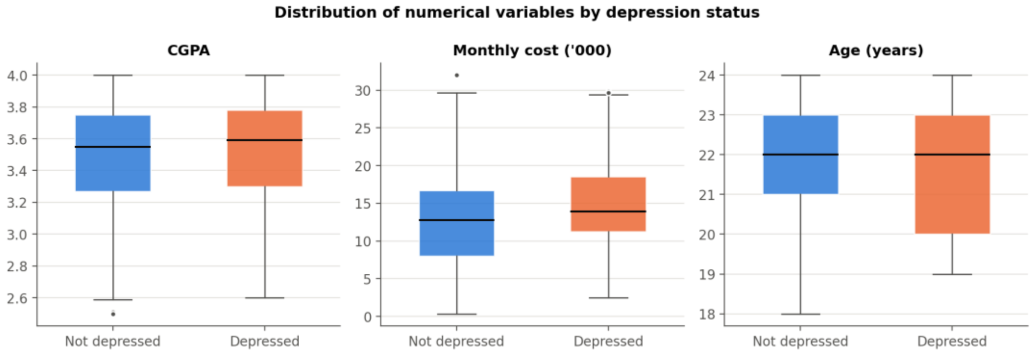


Fig. 4. Distribution of CGPA, monthly cost and age by depression status.

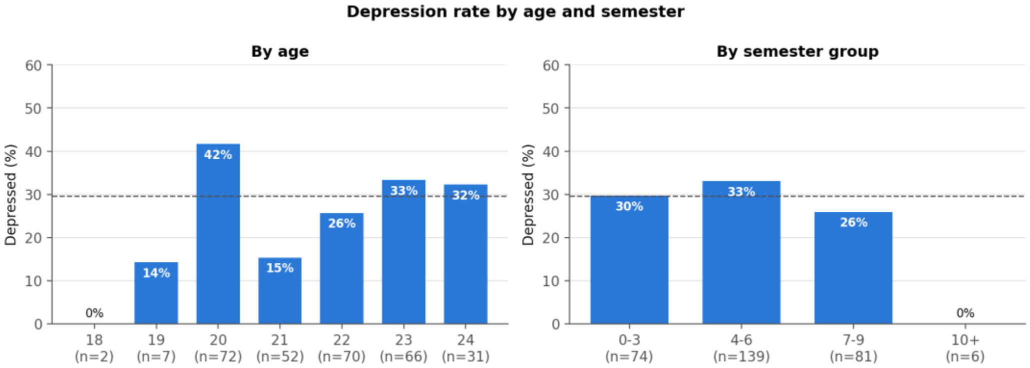


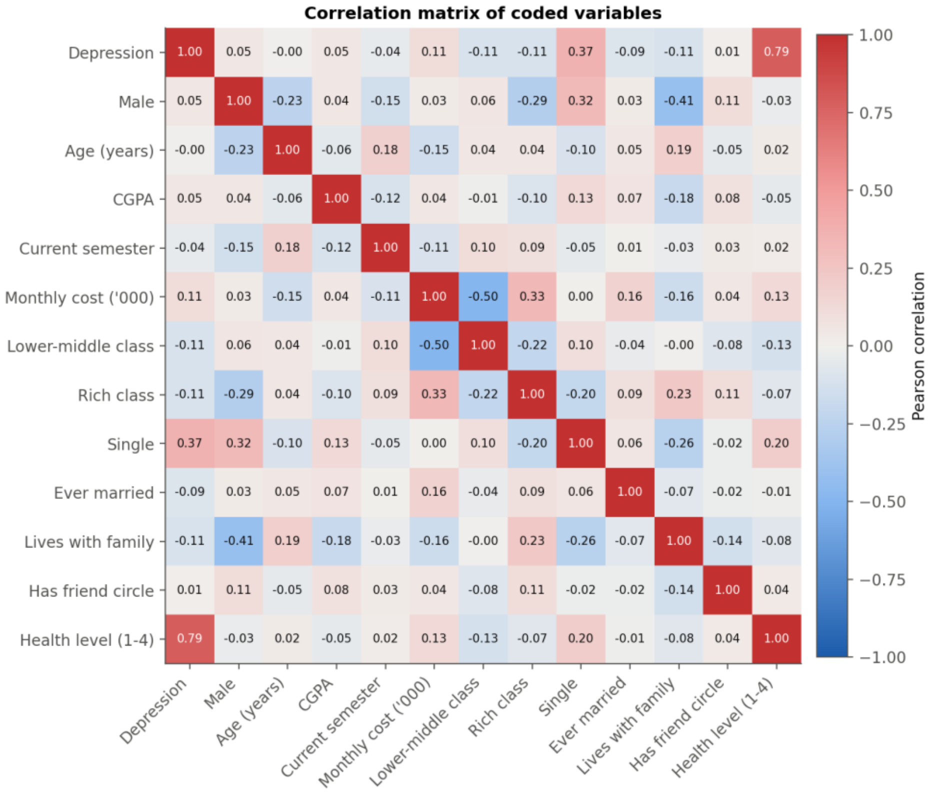
Fig. 5. Depression rate by age and semester group.

Table III reports summary statistics. Students are on average 21.7 years old with a mean CGPA of 3.49 and a mean monthly cost of 13.4 thousand; 64% are single, 71% have a friend circle and 39% live with family.

TABLE III
SUMMARY STATISTICS OF CODED VARIABLES

Variable	Obs	Mean	Std. dev.	Min	Max
Depression	300	0.297	0.458	0.000	1.000
Male	300	0.480	0.500	0.000	1.000
Age (years)	300	21.677	1.409	18.000	24.000
CGPA	300	3.486	0.352	2.500	4.000
Current semester	300	5.193	2.204	0.000	11.000
Monthly cost ('000)	300	13.398	6.182	0.342	32.000
Lower-middle class	300	0.190	0.393	0.000	1.000
Rich class	300	0.167	0.373	0.000	1.000
Single	300	0.640	0.481	0.000	1.000
Ever married	300	0.150	0.358	0.000	1.000
Lives with family	300	0.387	0.488	0.000	1.000
Has friend circle	300	0.710	0.455	0.000	1.000
Poor health	300	0.373	0.484	0.000	1.000

The correlation matrix (Fig. 6) shows modest correlations among the regressors; the largest are between monthly cost and lower-middle class (−0.50) and between male and living with family (−0.41). The very high correlation between depression and health level (0.79) confirms the separation issue.



	Depression	Male	Age (years)	CGPA	Current semester	Monthly cost ('000)	Lower-middle class	Rich class	Single	Ever married	Lives with family	Has friend circle	Health level (1-4)
Depression	1.00	0.05	-0.00	0.05	-0.04	0.11	-0.11	-0.11	0.37	-0.09	-0.11	0.01	0.79
Male	0.05	1.00	-0.23	0.04	-0.15	0.03	0.06	-0.29	0.32	0.03	-0.41	0.11	-0.03
Age (years)	-0.00	-0.23	1.00	-0.06	0.18	-0.15	0.04	0.04	-0.10	0.05	0.19	-0.05	0.02
CGPA	0.05	0.04	-0.06	1.00	-0.12	0.04	-0.01	-0.10	0.13	0.07	-0.18	0.08	-0.05
Current semester	-0.04	-0.15	0.18	-0.12	1.00	-0.11	0.10	0.09	-0.05	0.01	-0.03	0.03	0.02
Monthly cost ('000)	0.11	0.03	-0.15	0.04	-0.11	1.00	-0.50	0.33	0.00	0.16	-0.16	0.04	0.13
Lower-middle class	-0.11	0.06	0.04	-0.01	0.10	-0.50	1.00	-0.22	0.10	-0.04	-0.00	-0.08	-0.13
Rich class	-0.11	-0.29	0.04	-0.10	0.09	0.33	-0.22	1.00	-0.20	0.09	0.23	0.11	-0.07
Single	0.37	0.32	-0.10	0.13	-0.05	0.00	0.10	-0.20	1.00	0.06	-0.26	-0.02	0.20
Ever married	-0.09	0.03	0.05	0.07	0.01	0.16	-0.04	0.09	0.06	1.00	-0.07	-0.02	-0.01
Lives with family	-0.11	-0.41	0.19	-0.18	-0.03	-0.16	-0.00	0.23	-0.26	-0.07	1.00	-0.14	-0.08
Has friend circle	0.01	0.11	-0.05	0.08	0.03	0.04	-0.08	0.11	-0.02	-0.02	-0.14	1.00	0.04
Health level (1-4)	0.79	-0.03	0.02	-0.05	0.02	0.13	-0.13	-0.07	0.20	-0.01	-0.08	0.04	1.00

Fig. 6. Correlation matrix of coded variables.

B. Bivariate Tests

Table IV reports chi-square tests of independence. Relationship status (χ^2 = 40.39, p < 0.001, Cramér's V = 0.37) and family background (χ^2 = 9.62, p = 0.008) are significantly associated with depression, as is health (V = 0.84). Living style is borderline (p = 0.054); gender, marital status and friend circle are not significant. Table V shows that no numerical variable differs significantly at 5%; monthly cost comes closest (Welch p = 0.067, Mann–Whitney p = 0.051), while CGPA (p = 0.388) shows no difference.

TABLE IV
CHI-SQUARE TESTS OF INDEPENDENCE WITH DEPRESSION

Variable	Chi-square	df	p-value	Cramer's V	Min expected count	Fisher exact p (2x2)
Gender	0.689	1	0.407	0.048	42.720	0.449
Family background	9.623	2	0.008	0.179	14.830	
Relationship status	40.394	2	<0.001	0.367	3.260	
Marital status	2.408	2	0.300	0.090	1.780	
Living style	3.702	1	0.054	0.111	34.410	0.069
Friend circle	0.051	1	0.821	0.013	25.810	0.890
Health condition	212.060	3	<0.001	0.841	14.240	

TABLE V
DIFFERENCE IN MEANS BETWEEN DEPRESSED AND NON-DEPRESSED STUDENTS

Variable	Mean (not depressed)	Mean (depressed)	Difference	Welch t	t-test p-value	Mann-Whitney U	Mann-Whitney p-value
CGPA	3.475	3.513	0.038	0.865	0.388	10211.500	0.231
Monthly cost ('000)	12.960	14.434	1.473	1.848	0.067	10726.500	0.051
Age (years)	21.678	21.674	-0.004	-0.019	0.984	9312.000	0.909
Current semester	5.256	5.045	-0.211	-0.800	0.424	9060.500	0.628

C. Multicollinearity

The largest variance inflation factor is 1.61 (Table VI), far below the common threshold of 10 (and the stricter 5), so multicollinearity is not a concern.

TABLE VI
VARIANCE INFLATION FACTORS

Variable	VIF	Tolerance (1/VIF)
Male	1.410	0.709
Age (years)	1.127	0.887
CGPA	1.077	0.929
Current semester	1.110	0.901
Monthly cost ('000)	1.614	0.620
Lower-middle class	1.378	0.726
Rich class	1.371	0.729
Single	1.176	0.850
Ever married	1.053	0.950
Lives with family	1.405	0.712
Has friend circle	1.076	0.929

D. Binary Logit Model

Table VII presents the logit estimates. The model is highly significant (LR $\chi^2(11)$ = 69.57, p < 0.001) with McFadden's pseudo-R^2 = 0.191, a reasonable value for individual cross-sectional data.

TABLE VII
BINARY LOGIT ESTIMATES (N = 300; DEPENDENT VARIABLE: DEPRESSION)

Variable	Coefficient	Std. err.	z	P>|z|	95% CI lower	95% CI upper	Sig.
Constant	-5.1987	2.9172	-1.7821	0.0747	-10.9163	0.5190	*
Male	-0.6574	0.3472	-1.8936	0.0583	-1.3378	0.0230	*
Age (years)	0.1244	0.1061	1.1727	0.2409	-0.0835	0.3324	
CGPA	-0.0885	0.4290	-0.2062	0.8366	-0.9292	0.7523	
Current semester	-0.0373	0.0718	-0.5188	0.6039	-0.1781	0.1035	
Monthly cost ('000)	0.0699	0.0357	1.9584	0.0502	-0.0001	0.1398	*
Lower-middle class	-0.7071	0.4649	-1.5210	0.1283	-1.6183	0.2041	
Rich class	-1.2038	0.5899	-2.0405	0.0413	-2.3600	-0.0475	**
Single	2.5285	0.4285	5.9005	<0.001	1.6886	3.3684	***
Ever married	-1.1145	0.4516	-2.4681	0.0136	-1.9995	-0.2295	**
Lives with family	-0.2743	0.3726	-0.7361	0.4617	-1.0046	0.4561	
Has friend circle	0.1859	0.3323	0.5592	0.5760	-0.4655	0.8372	

Holding other factors constant, being single has the strongest effect (β = 2.529, p < 0.001). Ever being married (β = -1.115, p = 0.014) and a rich family (β = -1.204, p = 0.041) lower the probability; monthly cost raises it (β = 0.070, p = 0.050); male students are less likely to be depressed (β = -0.657, p = 0.058). CGPA, age, semester, lower-middle class, living with family and friend circle are not significant.

In odds-ratio terms (Table VIII, Fig. 7), single students have 12.5 times the odds of depression of students in a relationship (95% CI 5.4–29.0); rich students have 70% lower odds than middle-class students; ever-married students have 67% lower odds; and each thousand units of monthly cost raises the odds by 7.2%.

TABLE VIII
ODDS RATIOS FROM THE LOGIT MODEL

Variable	Odds ratio	95% CI lower	95% CI upper	P>|z|	Sig.
Constant	0.005	0.000	1.680	0.075	*
Male	0.518	0.262	1.023	0.058	*
Age (years)	1.133	0.920	1.394	0.241	
CGPA	0.915	0.395	2.122	0.837	
Current semester	0.963	0.837	1.109	0.604	
Monthly cost ('000)	1.072	1.000	1.150	0.050	*
Lower-middle class	0.493	0.198	1.226	0.128	
Rich class	0.300	0.094	0.954	0.041	**
Single	12.535	5.412	29.033	<0.001	***
Ever married	0.328	0.135	0.795	0.014	**
Lives with family	0.760	0.366	1.578	0.462	
Has friend circle	1.204	0.628	2.310	0.576	

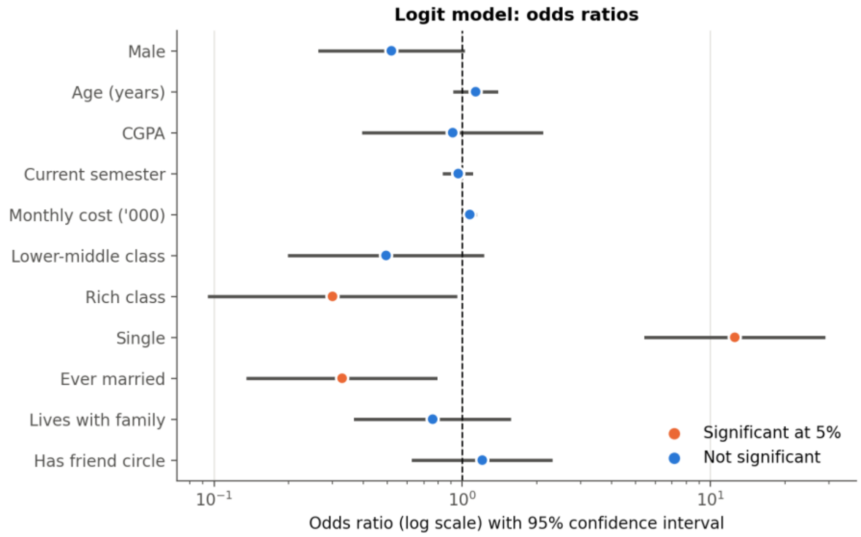


Fig. 7. Logit odds ratios with 95% confidence intervals (log scale).

E. Marginal Effects

Table IX and Fig. 8 report the average marginal effects. For a student with average characteristics the predicted probability of depression is 22.4%.

TABLE IX
AVERAGE MARGINAL EFFECTS AFTER LOGIT (DY/DX)

Variable	dy/dx	Std. err.	z	P>|z|	95% CI lower	95% CI upper	Sig.
Male	-0.1070	0.0545	-1.9633	0.0496	-0.2139	-0.0002	**
Age (years)	0.0205	0.0174	1.1825	0.2370	-0.0135	0.0546	
CGPA	-0.0146	0.0708	-0.2062	0.8366	-0.1533	0.1241	
Current semester	-0.0061	0.0118	-0.5196	0.6033	-0.0293	0.0170	
Monthly cost ('000)	0.0115	0.0058	2.0042	0.0451	0.0003	0.0228	**
Lower-middle class	-0.1118	0.0685	-1.6335	0.1024	-0.2460	0.0224	
Rich class	-0.1716	0.0671	-2.5587	0.0105	-0.3030	-0.0401	**
Single	0.3643	0.0425	8.5802	<0.001	0.2811	0.4475	***
Ever married	-0.1643	0.0557	-2.9497	0.0032	-0.2735	-0.0551	***
Lives with family	-0.0448	0.0600	-0.7468	0.4552	-0.1623	0.0727	
Has friend circle	0.0304	0.0538	0.5654	0.5718	-0.0750	0.1358	

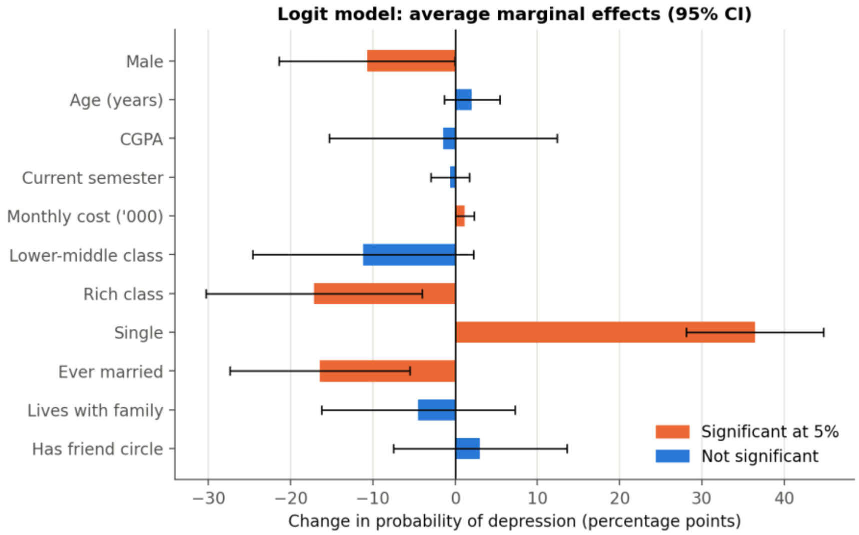


Fig. 8. Average marginal effects after logit with 95% confidence intervals.

- Being **single** raises the probability of depression by 36.4 percentage points (p < 0.001)—by far the largest effect.
- A **rich family** lowers it by 17.2 points relative to middle class (p = 0.011); **ever married** lowers it by 16.4 points (p = 0.003).
- Each thousand units of **monthly cost** adds 1.15 points (p = 0.045), about 12 points for a difference of 10 thousand.
- **Male** students are 10.7 points less likely to be depressed (p = 0.050).
- **CGPA** (-1.5 points per grade point, p = 0.837) and **friend circle** (p = 0.572) have no significant effect.

Marginal effects at the means (Stata's older mfx) lead to the same conclusions (see Appendix).

F. Binary Probit Model

The probit model (Table X) is equally significant (LR $\chi^2(11) = 69.87$, pseudo-$R^2 = 0.192$) with the same signs and significance pattern; the male coefficient is significant at 5% (p = 0.040). Its marginal effects (Table XI) are almost identical to the logit ones: single +36.8 percentage points (logit 36.4), rich -17.1 percentage points, ever married -14.9 percentage points. The choice of link function makes no practical difference.

TABLE X

BINARY PROBIT ESTIMATES (N = 300; DEPENDENT VARIABLE: DEPRESSION)

Variable	Coefficient	Std. err.	z	P>|z|	95% CI lower	95% CI upper	Sig.
Constant	-3.0986	1.7439	-1.7769	0.0756	-6.5166	0.3193	*
Male	-0.4204	0.2043	-2.0578	0.0396	-0.8207	-0.0200	**
Age (years)	0.0673	0.0630	1.0681	0.2855	-0.0562	0.1907	
CGPA	-0.0210	0.2538	-0.0828	0.9340	-0.5185	0.4765	
Current semester	-0.0159	0.0418	-0.3810	0.7032	-0.0978	0.0659	
Monthly cost ('000)	0.0436	0.0209	2.0833	0.0372	0.0026	0.0847	**
Lower-middle class	-0.3983	0.2750	-1.4486	0.1475	-0.9373	0.1406	
Rich class	-0.7000	0.3404	-2.0566	0.0397	-1.3671	-0.0329	**
Single	1.4643	0.2306	6.3495	<0.001	1.0123	1.9163	***
Ever married	-0.5894	0.2532	-2.3278	0.0199	-1.0856	-0.0931	**
Lives with family	-0.1586	0.2139	-0.7417	0.4583	-0.5778	0.2606	
Has friend circle	0.1344	0.1976	0.6802	0.4964	-0.2529	0.5217	

TABLE XI
AVERAGE MARGINAL EFFECTS AFTER PROBIT (DY/DX)

Variable	dy/dx	Std. err.	z	P>|z|	95% CI lower	95% CI upper	Sig.
Male	-0.1149	0.0537	-2.1397	0.0324	-0.2201	-0.0096	**
Age (years)	0.0187	0.0174	1.0731	0.2832	-0.0154	0.0527	
CGPA	-0.0058	0.0704	-0.0828	0.9340	-0.1439	0.1322	
Current semester	-0.0044	0.0116	-0.3810	0.7032	-0.0271	0.0183	
Monthly cost ('000)	0.0121	0.0057	2.1320	0.0330	0.0010	0.0232	**
Lower-middle class	-0.1060	0.0689	-1.5387	0.1239	-0.2411	0.0290	
Rich class	-0.1706	0.0680	-2.5074	0.0122	-0.3039	-0.0372	**
Single	0.3676	0.0426	8.6251	<0.001	0.2841	0.4512	***
Ever married	-0.1487	0.0560	-2.6560	0.0079	-0.2584	-0.0390	***
Lives with family	-0.0437	0.0582	-0.7499	0.4533	-0.1578	0.0705	
Has friend circle	0.0369	0.0535	0.6895	0.4905	-0.0680	0.1418	

G. Goodness of Fit

Both models fit well and almost equally (Table XII). The Hosmer–Lemeshow test does not reject the logit model ($\chi^2 = 9.34$, $p = 0.314$), so predicted and observed rates agree across risk groups. At a 0.5 cut-off the logit model classifies 73.3% of students correctly, with specificity 85.8% but sensitivity only 43.8% (Table XIII). Using the sample share of depressed students (0.297) as cut-off identifies 79.8% of depressed students while correctly classifying 65.4% of the others.

TABLE XII
GOODNESS-OF-FIT STATISTICS

Statistic	Logit	Probit
Observations	300.000	300.000
Log likelihood	-147.619	-147.471
Log likelihood (null)	-182.404	-182.404
LR chi2	69.571	69.865
LR df	11.000	11.000
Prob > chi2	0.000	0.000
Pseudo R2 (McFadden)	0.191	0.192
AIC	319.237	318.943
BIC	363.683	363.388
AUC (ROC)	0.781	0.781
Hosmer-Lemeshow chi2	9.343	12.160
HL df	8.000	8.000
HL p-value	0.314	0.144

TABLE XIII
CLASSIFICATION TABLES (CUT-OFF 0.5 AND SAMPLE SHARE)

Model	Cutoff	True positives	False positives	False negatives	True negatives	Sensitivity	Specificity	Correctly classified
Logit	0.500	39	30	50	181	0.438	0.858	0.733
Logit	0.297	71	73	18	138	0.798	0.654	0.697
Probit	0.500	37	28	52	183	0.416	0.867	0.733
Probit	0.297	74	74	15	137	0.832	0.649	0.703

The ROC curves (Fig. 9) give an AUC of 0.781 (logit) and 0.781 (probit): a randomly chosen depressed student has a 78% chance of a higher predicted probability than a randomly chosen non-depressed student. Fig. 10 shows the size of the relationship effect: with other variables at their means, the predicted probability is above 40% for single students across the whole CGPA range versus about 5–6% for other students, and the nearly flat lines confirm that CGPA matters little.

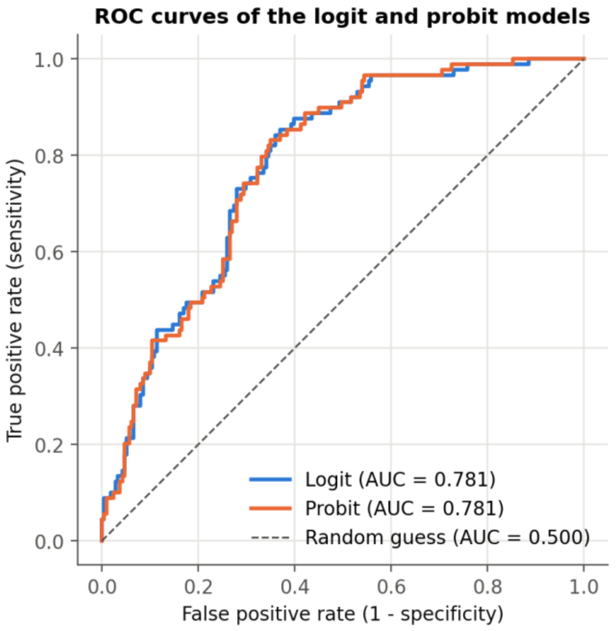


Fig. 9. ROC curves of the logit and probit models.

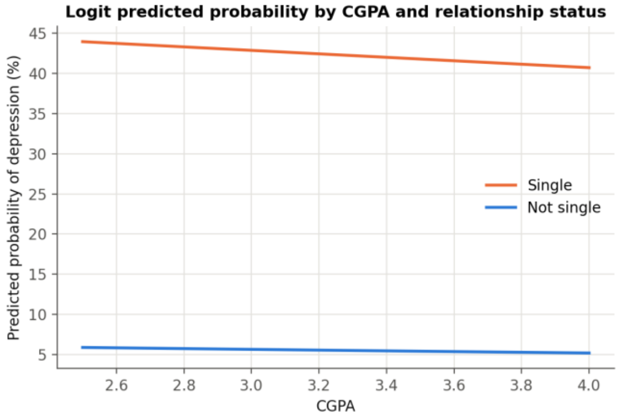


Fig. 10. Predicted probability of depression by CGPA and relationship status.

H. Hypothesis Tests

Table XIV summarises the hypothesis tests (logit model; supported if the expected sign holds with $p < 0.05$). H3 and H4b are supported; H5 and H7 are supported at 10% in the logit coefficients and at 5% in their marginal effects and in the probit model. H1, H2, H4a, H6 are not supported: academic performance, friend circle, lower-middle class background and living with family have no significant independent effect.

TABLE XIV
SUMMARY OF HYPOTHESIS TESTS

Hypothesis	Variable	Expected sign	Logit coefficient	AME	P>|z|	Result
H1	CGPA	-	-0.0885	-0.0146	0.8366	Not supported (not significant)
H2	Has friend circle	-	0.1859	0.0304	0.5760	Not supported (not significant)
H3	Single	+	2.5285	0.3643	<0.001	Supported
H4a	Lower-middle class	+	-0.7071	-0.1118	0.1283	Not supported (not significant)
H4b	Rich class	-	-1.2038	-0.1716	0.0413	Supported
H5	Monthly cost ('000)	+	0.0699	0.0115	0.0502	Weakly supported (10% level)
H6	Lives with family	-	-0.2743	-0.0448	0.4617	Not supported (not significant)
H7	Male	-	-0.6574	-0.1070	0.0583	Weakly supported (10% level)

I. Robustness Checks

Table XV compares the main model with six alternatives, and Fig. 11 compares the marginal effects of the logit, probit and linear probability models, which agree closely on every variable.

TABLE XV
ROBUSTNESS CHECKS ACROSS MODEL SPECIFICATIONS

Variable	1 Logit	2 Logit rob.	3 Probit	4 LPM rob.	5 Logit+health	6 LPM male	7 LPM female
Constant	-5.199*	-5.199*	-3.099*	-0.297	-7.915	-0.219	-0.521
	(2.917)	(2.757)	(1.744)	(0.477)	(6.957)	(0.690)	(0.752)
Male	-0.657*	-0.657*	-0.420**	-0.095*	-1.153		
	(0.347)	(0.340)	(0.204)	(0.057)	(0.823)		
Age (years)	0.124	0.124	0.067	0.018	-0.247	-0.004	0.045
	(0.106)	(0.104)	(0.063)	(0.019)	(0.253)	(0.029)	(0.030)
CGPA	-0.088	-0.088	-0.021	-0.011	1.477	0.097	-0.109
	(0.429)	(0.432)	(0.254)	(0.070)	(1.016)	(0.123)	(0.096)
Current semester	-0.037	-0.037	-0.016	-0.003	0.080	0.005	-0.013
	(0.072)	(0.075)	(0.042)	(0.012)	(0.163)	(0.020)	(0.015)
Monthly cost ('000)	0.070*	0.070**	0.044**	0.009*	0.001	0.004	0.010*
	(0.036)	(0.033)	(0.021)	(0.005)	(0.069)	(0.011)	(0.005)
Lower-middle class	-0.707	-0.707	-0.398	-0.131*	-0.831	-0.234**	-0.013
	(0.465)	(0.441)	(0.275)	(0.068)	(1.126)	(0.113)	(0.099)
Rich class	-1.204**	-1.204**	-0.700**	-0.139**	0.868	-0.322**	-0.101
	(0.590)	(0.511)	(0.340)	(0.061)	(1.053)	(0.159)	(0.077)
Single	2.529***	2.529***	1.464***	0.376***	5.356***	0.390***	0.320***
	(0.429)	(0.413)	(0.231)	(0.049)	(1.004)	(0.071)	(0.070)
Ever married	-1.115**	-1.115**	-0.589**	-0.163**	-2.710***	-0.251***	-0.049
	(0.452)	(0.505)	(0.253)	(0.071)	(1.045)	(0.093)	(0.118)
Lives with family	-0.274	-0.274	-0.159	-0.022	-1.058	0.028	-0.028
	(0.373)	(0.378)	(0.214)	(0.057)	(0.770)	(0.091)	(0.073)
Has friend circle	0.186	0.186	0.134	0.029	-0.636	-0.004	0.013
	(0.332)	(0.317)	(0.198)	(0.053)	(0.714)	(0.085)	(0.074)
Poor health					7.958***		
					(1.122)		
Observations	300	300	300	300	300	144	156
Pseudo R2 / R2	0.191	0.191	0.192	0.192	0.791	0.233	0.202

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors in parentheses. Models 6–7 are linear probability models with robust standard errors.

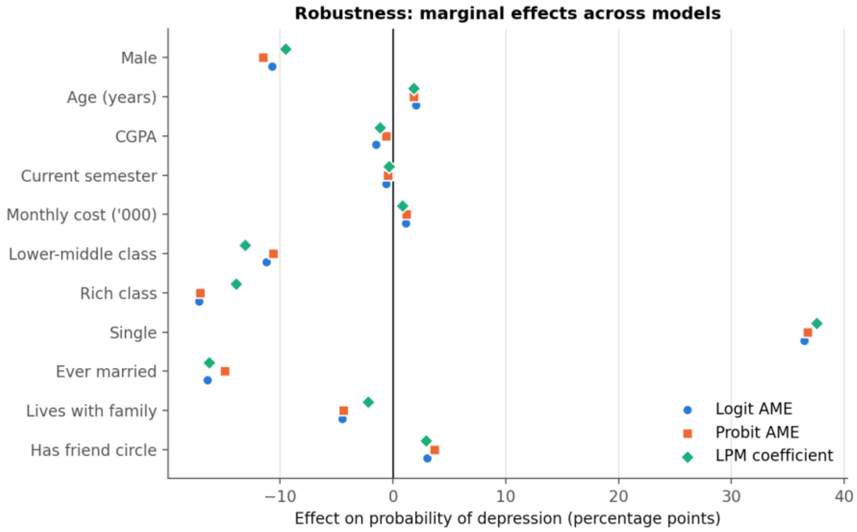


Fig. 11. Marginal effects across logit, probit and linear probability models.

- **Robust standard errors** change the standard errors only slightly; every variable keeps its sign and significance level, except monthly cost, which becomes significant at 5%.
- **The LPM** coefficients, which are marginal effects directly, are very close to the logit and probit AMEs ($R^2 = 0.192$).
- **Adding poor health** raises the pseudo-R^2 from 0.191 to 0.791 and the AUC to 0.986 (LR $\chi^2(1) = 218.8$, $p < 0.001$); the software flags quasi-complete separation (a fraction 0.12 of observations predicted perfectly), confirming that health largely measures the outcome itself. Single and ever married remain significant.
- **By gender,** a logit cannot be estimated for men because none of the 29 non-single male students is depressed, so LPMs are used. Being single is a strong risk factor for men (+39.0 percentage points, $p < 0.001$) and women (+32.0 percentage points, $p < 0.001$). Rich family (-32.2 percentage points), ever married (-25.1 percentage points) and lower-middle class (-23.4 percentage points) are protective among men but not significant among women ($p = 0.19$, 0.68, 0.89); monthly cost matters mainly for women (+1.03 points per thousand, $p = 0.058$). With fewer than 50 depressed students per subsample, these differences are indicative.

V. DISCUSSION

A. Key Findings

Relationships matter more than grades. Single students are 36 percentage points more likely to be depressed, an effect that is large, highly significant and stable across every specification and for both men and women; ever-married students are less likely to be depressed. This strongly supports the stress-buffering view of close relationships [5].

The 'squeezed middle'. Middle-class students are the most likely to be depressed, and after controls rich students are 17 points less likely to be. Higher living costs raise the probability. Middle-class students may face high expectations and rising costs without the cushion of richer families, consistent with research on financial strain [10].

The gender gap. After controls, female students are about 11 points more likely to be depressed. The gap is hidden in the raw data—where men show a slightly higher rate—because male students are more often single. The adjusted result matches the established gender difference [7], [13].

What did not matter. CGPA, semester and age have no significant relationship with depression, and neither do a friend circle or living with family. Living with family looks protective in the raw data, but the effect disappears once relationship status and gender are controlled. The support that matters most here seems to be an intimate relationship rather than a wider friend group.

B. Implications for Universities

Since grades do not predict depression in this sample, services focused mainly on academic difficulty will miss many depressed students. Universities should (i) invest in programmes that reduce loneliness and build close connections—peer mentoring, small-group and residential community activities; (ii) combine mental health support with financial guidance and need-based help that also reaches middle-income students; (iii) design gender-sensitive outreach, since women face higher adjusted risk while men may be less likely to seek help; and (iv) use simple risk scores to prioritise screening—the logit model identifies 80% of depressed students at the sample-share cut-off—without replacing clinical assessment.

C. Theoretical Implications

The results support the diathesis–stress and stress-buffering frameworks [4], [5]: financial pressure raises the probability of depression while close relationships protect against it. They suggest that for young adults the protective effect comes from intimate relationships more than from the size of the social network, and the non-linear pattern of family background suggests that relative pressure and expectations may matter as much as absolute income.

VI. LIMITATIONS AND FUTURE RESEARCH

The data are cross-sectional, so the estimates are associations rather than causal effects; depression may itself affect relationship status. Depression and health are self-reported indicators rather than clinical diagnoses or validated scales. With 300 students and 89 depressed cases, small categories had to be merged and the gender subsample results are indicative. The sample comes from one region, which removes regional confounding but limits generalisation, and factors such as family history, sleep, physical activity and substance use are not observed. Future work should follow students over several semesters, use validated severity scales such as the PHQ-9 with ordered models, draw larger multi-region samples, and measure the quality—not only the presence—of social support.

VII. CONCLUSION

Depression affects almost three in ten students in this sample. It is driven less by grades and more by students' relationships and financial circumstances: being single is the strongest risk factor, while being married, coming from a rich family and being male are associated with lower risk, and higher living costs raise the risk. These findings are consistent across logit, probit and linear probability models. Protecting student mental health therefore requires attention to social connection and financial wellbeing, not only to academic support.

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APPENDIX: FULL MODEL OUTPUT

Full estimation output from the analysis notebook (notebooks/student_depression_analysis.ipynb, Python statsmodels). The Stata do-file stata/student_depression_analysis.do contains the equivalent commands (summarize, tabulate, logit, probit, margins, estat classification, estat gof, lroc, regress). All tables are also in results/Student_Depression_Analysis_Results.xlsx.

A. Binary logit model

```
                           Logit Regression Results
==============================================================================
Dep. Variable:             depression   No. Observations:                  300
Model:                          Logit   Df Residuals:                      288
Method:                           MLE   Df Model:                           11
Date:                Tue, 06 Oct 2026   Pseudo R-squ.:                  0.1907
Time:                        16:44:00   Log-Likelihood:                -147.62
converged:                       True   LL-Null:                       -182.40
Covariance Type:            nonrobust   LLR p-value:                 1.474e-10
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -5.1987      2.917     -1.782      0.075     -10.916       0.519
male             -0.6574      0.347     -1.894      0.058      -1.338       0.023
age               0.1244      0.106      1.173      0.241      -0.084       0.332
cgpa             -0.0885      0.429     -0.206      0.837      -0.929       0.752
semester         -0.0373      0.072     -0.519      0.604      -0.178       0.104
cost_k            0.0699      0.036      1.958      0.050    -5.5e-05       0.140
fam_lowermid     -0.7071      0.465     -1.521      0.128      -1.618       0.204
fam_rich         -1.2038      0.590     -2.041      0.041      -2.360      -0.048
single            2.5285      0.429      5.900      0.000       1.689       3.368
ever_married     -1.1145      0.452     -2.468      0.014      -2.000      -0.229
live_family      -0.2743      0.373     -0.736      0.462      -1.005       0.456
friend_circle     0.1859      0.332      0.559      0.576      -0.466       0.837
```

```
=================================================================================
```

B. Binary probit model

```
                          Probit Regression Results
==============================================================================
Dep. Variable:             depression   No. Observations:                  300
Model:                         Probit   Df Residuals:                      288
Method:                           MLE   Df Model:                           11
Date:                Tue, 06 Oct 2026   Pseudo R-squ.:                  0.1915
Time:                        16:44:01   Log-Likelihood:                -147.47
converged:                       True   LL-Null:                       -182.40
Covariance Type:            nonrobust   LLR p-value:                 1.296e-10
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -3.0986      1.744     -1.777      0.076      -6.517       0.319
male             -0.4204      0.204     -2.058      0.040      -0.821      -0.020
age               0.0673      0.063      1.068      0.285      -0.056       0.191
cgpa             -0.0210      0.254     -0.083      0.934      -0.519       0.476
semester         -0.0159      0.042     -0.381      0.703      -0.098       0.066
cost_k            0.0436      0.021      2.083      0.037       0.003       0.085
fam_lowermid     -0.3983      0.275     -1.449      0.147      -0.937       0.141
fam_rich         -0.7000      0.340     -2.057      0.040      -1.367      -0.033
single            1.4643      0.231      6.349      0.000       1.012       1.916
ever_married     -0.5894      0.253     -2.328      0.020      -1.086      -0.093
live_family      -0.1586      0.214     -0.742      0.458      -0.578       0.261
friend_circle     0.1344      0.198      0.680      0.496      -0.253       0.522
=================================================================================
```

C. Logit model with robust standard errors

```
                           Logit Regression Results
==============================================================================
Dep. Variable:             depression   No. Observations:                  300
Model:                          Logit   Df Residuals:                      288
Method:                           MLE   Df Model:                           11
Date:                Tue, 06 Oct 2026   Pseudo R-squ.:                  0.1907
Time:                        16:44:01   Log-Likelihood:                -147.62
converged:                       True   LL-Null:                       -182.40
Covariance Type:                  HC1   LLR p-value:                 1.474e-10
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -5.1987      2.757     -1.885      0.059     -10.603       0.206
male             -0.6574      0.340     -1.931      0.053      -1.325       0.010
age               0.1244      0.104      1.200      0.230      -0.079       0.328
cgpa             -0.0885      0.432     -0.205      0.838      -0.935       0.758
semester         -0.0373      0.075     -0.499      0.618      -0.184       0.109
cost_k            0.0699      0.033      2.138      0.033       0.006       0.134
fam_lowermid     -0.7071      0.441     -1.602      0.109      -1.572       0.158
fam_rich         -1.2038      0.511     -2.356      0.018      -2.205      -0.202
single            2.5285      0.413      6.126      0.000       1.720       3.337
ever_married     -1.1145      0.505     -2.207      0.027      -2.104      -0.125
live_family      -0.2743      0.378     -0.725      0.468      -1.016       0.467
friend_circle     0.1859      0.317      0.586      0.558      -0.435       0.807
=================================================================================
```

D. Linear probability model (robust standard errors)

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             depression   R-squared:                       0.192
Model:                            OLS   Adj. R-squared:                  0.161
Method:                 Least Squares   F-statistic:                     9.377
Date:                Tue, 06 Oct 2026   Prob (F-statistic):           2.09e-14
Time:                        16:44:01   Log-Likelihood:                -158.73
No. Observations:                 300   AIC:                             341.5
Df Residuals:                     288   BIC:                             385.9
Df Model:                          11
Covariance Type:                  HC1
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -0.2966      0.477     -0.622      0.534      -1.231       0.638
male             -0.0952      0.057     -1.656      0.098      -0.208       0.017
age               0.0185      0.019      0.996      0.319      -0.018       0.055
cgpa             -0.0113      0.070     -0.162      0.871      -0.148       0.125
semester         -0.0033      0.012     -0.284      0.776      -0.026       0.019
cost_k            0.0085      0.005      1.822      0.068      -0.001       0.018
fam_lowermid     -0.1307      0.068     -1.929      0.054      -0.264       0.002
fam_rich         -0.1387      0.061     -2.261      0.024      -0.259      -0.018
single            0.3755      0.049      7.670      0.000       0.280       0.471
ever_married     -0.1625      0.071     -2.292      0.022      -0.301      -0.024
live_family      -0.0219      0.057     -0.384      0.701      -0.134       0.090
friend_circle     0.0295      0.053      0.553      0.580      -0.075       0.134
==============================================================================
Omnibus:                       86.457   Durbin-Watson:                   1.936
```

```
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               26.568
Skew:                           0.500   Prob(JB):                     1.70e-06
Kurtosis:                       1.939   Cond. No.                         540.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

E. Logit model including poor health

```
                           Logit Regression Results
==============================================================================
Dep. Variable:             depression   No. Observations:                  300
Model:                          Logit   Df Residuals:                      287
Method:                           MLE   Df Model:                           12
Date:                Tue, 06 Oct 2026   Pseudo R-squ.:                  0.7905
Time:                        16:44:01   Log-Likelihood:                -38.209
converged:                       True   LL-Null:                       -182.40
Covariance Type:            nonrobust   LLR p-value:                 1.281e-54
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -7.9152      6.957     -1.138      0.255     -21.550       5.720
male             -1.1528      0.823     -1.400      0.162      -2.767       0.461
age              -0.2471      0.253     -0.978      0.328      -0.742       0.248
cgpa              1.4769      1.016      1.453      0.146      -0.515       3.468
semester          0.0799      0.163      0.489      0.625      -0.240       0.400
cost_k            0.0009      0.069      0.013      0.990      -0.135       0.137
fam_lowermid     -0.8308      1.126     -0.738      0.460      -3.037       1.375
fam_rich          0.8678      1.053      0.824      0.410      -1.196       2.931
single            5.3555      1.004      5.333      0.000       3.387       7.324
ever_married     -2.7102      1.045     -2.594      0.009      -4.758      -0.663
live_family      -1.0582      0.770     -1.375      0.169      -2.567       0.450
friend_circle    -0.6355      0.714     -0.890      0.374      -2.035       0.764
health_poor       7.9576      1.122      7.091      0.000       5.758      10.157
=================================================================================

Possibly complete quasi-separation: A fraction 0.12 of observations can be
perfectly predicted. This might indicate that there is complete
quasi-separation. In this case some parameters will not be identified.
```

F. Linear probability model: male students

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             depression   R-squared:                       0.233
Model:                            OLS   Adj. R-squared:                  0.175
Method:                 Least Squares   F-statistic:                     6.216
Date:                Tue, 06 Oct 2026   Prob (F-statistic):           9.33e-08
Time:                        16:44:02   Log-Likelihood:                -75.366
No. Observations:                 144   AIC:                             172.7
Df Residuals:                     133   BIC:                             205.4
Df Model:                          10
Covariance Type:                  HC1
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -0.2187      0.690     -0.317      0.751      -1.570       1.133
age              -0.0040      0.029     -0.137      0.891      -0.061       0.053
cgpa              0.0966      0.123      0.786      0.432      -0.144       0.337
semester          0.0050      0.020      0.252      0.801      -0.034       0.044
cost_k            0.0041      0.011      0.372      0.710      -0.017       0.026
fam_lowermid     -0.2336      0.113     -2.058      0.040      -0.456      -0.011
fam_rich         -0.3216      0.159     -2.019      0.043      -0.634      -0.009
single            0.3902      0.071      5.526      0.000       0.252       0.529
ever_married     -0.2512      0.093     -2.706      0.007      -0.433      -0.069
live_family       0.0276      0.091      0.305      0.761      -0.150       0.205
friend_circle    -0.0045      0.085     -0.053      0.958      -0.170       0.161
==============================================================================
Omnibus:                       55.260   Durbin-Watson:                   1.945
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               11.188
Skew:                           0.343   Prob(JB):                      0.00372
Kurtosis:                       1.819   Cond. No.                         523.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

G. Linear probability model: female students

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             depression   R-squared:                       0.202
Model:                            OLS   Adj. R-squared:                  0.147
Method:                 Least Squares   F-statistic:                     5.236
Date:                Tue, 06 Oct 2026   Prob (F-statistic):           1.50e-06
Time:                        16:44:02   Log-Likelihood:                -78.048
No. Observations:                 156   AIC:                             178.1
```

```
Df Residuals:                     145   BIC:                             211.6
Df Model:                          10
Covariance Type:                  HC1
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -0.5214      0.752     -0.693      0.488      -1.996       0.953
age               0.0453      0.030      1.526      0.127      -0.013       0.103
cgpa             -0.1085      0.096     -1.131      0.258      -0.296       0.079
semester         -0.0133      0.015     -0.873      0.383      -0.043       0.017
cost_k            0.0103      0.005      1.895      0.058      -0.000       0.021
fam_lowermid     -0.0134      0.099     -0.135      0.893      -0.208       0.181
fam_rich         -0.1006      0.077     -1.306      0.192      -0.252       0.050
single            0.3199      0.070      4.557      0.000       0.182       0.457
ever_married     -0.0485      0.118     -0.413      0.680      -0.279       0.182
live_family      -0.0276      0.073     -0.377      0.706      -0.171       0.116
friend_circle     0.0126      0.074      0.169      0.866      -0.133       0.159
==============================================================================
Omnibus:                       17.043   Durbin-Watson:                   1.679
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               13.073
Skew:                           0.602   Prob(JB):                      0.00145
Kurtosis:                       2.251   Cond. No.                         591.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)
```

Table XVI
Marginal Effects at the Means after Logit

Variable	dy/dx	Std. err.	z	P>|z|	95% CI lower	95% CI upper	Sig.
Male	-0.1135	0.0589	-1.9279	0.0539	-0.2288	0.0019	*
Age (years)	0.0217	0.0184	1.1773	0.2391	-0.0144	0.0577	
CGPA	-0.0154	0.0746	-0.2063	0.8366	-0.1616	0.1308	
Current semester	-0.0065	0.0125	-0.5197	0.6032	-0.0309	0.0180	
Monthly cost ('000)	0.0122	0.0061	1.9975	0.0458	0.0002	0.0241	**
Lower-middle class	-0.1083	0.0626	-1.7309	0.0835	-0.2310	0.0143	*
Rich class	-0.1653	0.0599	-2.7598	0.0058	-0.2826	-0.0479	***
Single	0.3639	0.0443	8.2150	<0.001	0.2771	0.4508	***
Ever married	-0.1539	0.0482	-3.1933	0.0014	-0.2484	-0.0594	***
Lives with family	-0.0469	0.0623	-0.7531	0.4514	-0.1690	0.0752	
Has friend circle	0.0316	0.0554	0.5716	0.5676	-0.0769	0.1401	

Table XVII
Marginal Effects for Male and Female Students (LPM, Robust SE)

Variable	Female students: dy/dx	Male students: dy/dx	Female students: P>|z|	Male students: P>|z|
Age (years)	0.0453	-0.0040	0.1271	0.8910
CGPA	-0.1085	0.0966	0.2579	0.4319
Current semester	-0.0133	0.0050	0.3827	0.8011
Ever married	-0.0485	-0.2512	0.6799	0.0068
Has friend circle	0.0126	-0.0045	0.8660	0.9576
Lives with family	-0.0276	0.0276	0.7065	0.7607
Lower-middle class	-0.0134	-0.2336	0.8927	0.0396
Monthly cost ('000)	0.0103	0.0041	0.0581	0.7096
Rich class	-0.1006	-0.3216	0.1917	0.0435
Single	0.3199	0.3902	<0.001	<0.001